MTH313M PRACTICE PROBLEMS WEEK 3

Question 1. Let $X_t = \sum_{j=1}^{N_t} Y_j$ be a compound PP with rate $\lambda > 0$ and component distribution F (i.e. $Y_1, Y_2, \cdots$ are i.i.d. with common DF F). Verify that the Characteristic function of X_t is given by

$$\mathbb{E}e^{iuX_t} = e^{-\lambda t(1-\phi(u))}, u \in \mathbb{R}$$

where ϕ denotes the Characteristic function of Y_1.

Question 2. Assume that

(i) Shocks appear in a system according to a PP with rate $\lambda > 0$

(ii) Shocks cause indepedent and identically distributed damages $Y_1, Y_2, \cdots$ which are Exponentially distributed with mean $\frac{1}{\mu}$ for some $\mu > 0$

(iii) the damages accumulates additively and once the total damage exceeds a known threshold $\alpha > 0$, the system fails

(iv) the system continues to work till the total damage is less than α

If T denotes the time of failure of the system, find $\mathbb{E}T$.

Question 3. Suppose that the number of trials to be performed follows a Poisson distribution with mean $\lambda > 0$. Each trial has n possible outcomes, and independent of everything else, results in outcome i with probability p_i (of course, $\sum_i p_i = 1$). Let X_j denote the number of outcomes that occur exactly j times. Find $\mathbb{E}X_j$ and $Var(X_j)$.

Question 4. Complete the exercises left out in class.